

Continuous Assessment Test – I

Regulations – R2021

Degree & Branch	BE CSE				Semester	V
Subject Code & Name	UCS 2501 Computer Networks					
Academic Year	2023-2024 ODD	Batch	2021-2025	Date	20.09.2023	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Describe the principles, design, terminology and concepts of the network models (K2)
CO2:	Select the protocols of data link layer and apply error control mechanisms (K3)
CO3:	Choose the core functions of network layer protocols and apply them for data communication (K3)
CO4:	Select the Transport Layer Protocols suitable for transmission of data. (K3)
CO5:	Illustrate the Application Layer Protocols and their basic functionalities (K2).

Part – A ($4 \times 2 = 8$ Marks)

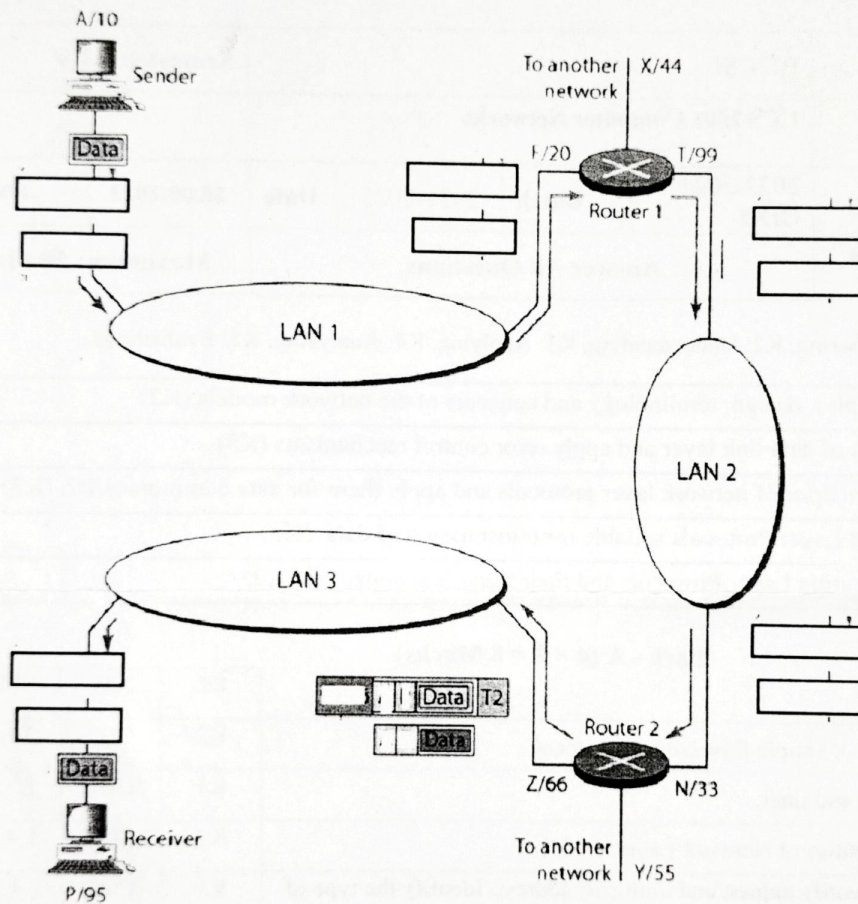
		KL	CO	PI
1.	Illustrate the FSM of simple flow control protocol	K2	CO2	2.1.2
2.	Define throughput and jitter.	K1	CO1	1.3.1
3.	State the functionalities of Network Layer.	K1	CO3	1.4.1
4.	State the rule to identify unicast and multicast address. Identify the type of address: C8:C2:A1:11:22:A2.	K3	CO2	2.4.1, 13.3.1

Part – B (3×6 = 18 Marks)

		KL	CO	PI
5.	Station A wants to send bit 1, station B sends bit 0, station C sends bit 1 and station D remains silent. These stations use CDMA MAC protocol to access the medium. Make use of CDMA and find the data on the common channel. Also, explain the procedure of CDMA.	K3	CO2	1.1.2, 1.3.1, 13.3.1
6.	Discuss about circuit switching and packet switching techniques with examples.	K2	CO3	1.1.2, 2.1.3, 2.3.1

7. Sender with IP address "A" from port 10 sends data to the receiver with IP address "P" on port 95. Make use of protocol layering and complete the frame format of data link layer and network layer of all devices in the entire path using the following network. Also, explain the functionalities of all layers of TCP/IP suite. With the help of TCP/IP, elaborate on how data gets transmitted from the sender to receiver through the intermediate devices.

K3 CO2 1.4
13.3



Part – C ($2 \times 12 = 24$ Marks)
(Any subdivisions should be either 7+5 or 8+4)

		KL	CO	PI
8.	Explain the four topologies of computer networks with their advantages and disadvantages.	K2	CO1	1.3.1, 1.4.1, 13.3.1
(Or)				
9.	Explain the different types of transmission media with their benefits	K2	CO1	1.3.1, 1.4.1, 13.3.1
10.	Identify the error correction technique and make use of it to encode the given data: 1010101 Find the redundant bits for encoding. During transmission, last LSB is corrupted. On the receiving end, detect this error and correct it.	K3	CO2	2.1.2, 2.1.3, 2.3.1, 13.3.1
(Or)				
11.	Make use of HDLC to transmit data from Node X to Node Y with piggybacking. Identify suitable frames for transmission. • Data frames 0,1 are transmitted from X to Y. • Data frames 0,1,2 are sent from Y to X and ends its transfer. • Data frames 2,3,4 are sent from X to Y out of which 3 gets lost. • Y replies with acknowledgement • X retransmits the lost packet and ends transfer Draw the flow diagram with HDLC frame format structure and explain it.	K3	CO2	2.1.2, 2.1.3, 2.3.1, 13.3.1